Slide 1: AI-Powered Real Estate Property Valuation Assistant

Prodapt Capstone Project • Panel Presentation (8 Min Demo + 2 Min Q&A;)

- **Project Domain**: Real Estate & PropTech Valuation Intelligence
- **Presenter**: Senior ML/Software Engineer
- **Architecture**: Autonomous Multi-Agent Collaborative System with Hybrid RAG & Supervised ML
- **Evaluation**: Verified with DeepEval Recommendation Quality Benchmarks

Slide 2: Problem Statement & Business Need

Why Real Estate Appraisals Require GenAI Automation

- **Manual Search Inefficiencies**: Traditional appraisal requires manual scanning of thousands of historical records.
- **Valuation Inconsistency**: Heuristic condition and age adjustments introduce subjective bias.
- **Unquantified Market Risk**: Properties with deferred maintenance lack quantified financial risk scoring.
- **Business Objective**: Build an autonomous multi-agent valuation assistant delivering instant comp retrieval, ML predictions, risk scoring, and LLM-as-judge audit.

Slide 3: Solution Overview & High-Level Architecture

End-to-End Flow from Dataset to Executive Dossier

- **1. Data Layer**: Standardizes Ames Housing Dataset (2,930 records, 15 key fields, 1-10 quality scale).
- **2. Hybrid RAG**: Dense embeddings (all-MiniLM-L6-v2) + lexical TF-IDF keyword search.
- **3. Supervised ML**: Gradient Boosting Regressor (Test $R^2 = 0.9031$, MAPE = 8.46%).
- **4. Multi-Agent Core**: Classification, Valuation, Risk Assessment, and LLM-as-Judge Auditor.
- **5. Presentation**: FastAPI REST microservice + Streamlit interactive dashboard.

Slide 4: Hybrid RAG & Comparable-Property Prioritization

Ensuring Precise, Grounded Comparable Selection

- **Dense Embeddings**: 384-dimensional vectors with local caching for instant sub-5ms cosine retrieval.
- **Lexical Match**: Captures exact specs (bedrooms, quality grades, subdivision names).
- **Metadata Filtering**: Filters candidates on location, age, and square footage.
- **Prioritization Formula**: Multi-factor index weighting Location (30%), Living Area (25%), Quality (20%), Age (15%), and Room Count (10%).

Slide 5: Autonomous Multi-Agent Collaboration Workflow

Orchestrating Specialized Agents for Appraisal Precision

- **Guardrail Agent**: Physical boundary validation (100-15k sqft, 1-10 qual) & injection filter.
- **Classification Agent**: Market Tier, Typology, Condition Grade, and Target Demographics.
- **Valuation Agent**: Comp-median, mean, and size-weighted PPSF fair market value.
- **Risk Agent**: Quantifies age depreciation, condition gap, and market volatility (0-100 score).
- **LLM-as-Judge Agent**: Independent audit against ML regression and comps issuing verdicts (APPROVED).

Slide 6: Supervised ML Pricing & Feature Impact

Predictive Modeling Benchmark & What-If Simulation

- **Gradient Boosting Regressor**: Achieved Test R^2 = 0.9031, RMSE = $27,870, MAE = $15,889, MAPE = 8.46%.
- **Top Drivers**: Overall Quality (54%), Living Area (22%), Basement (8%), and Location Premium.
- **Location Premiums**: Stone Brook (+85.9% premium), Northridge (+89.1%), Edwards (-22.5%).
- **What-If Renovation Simulator**: Real-time simulation of additions, quality upgrades, and remodels.

Slide 7: Evaluation Results (DeepEval Framework)

Honest Dual-Mode Quality Framework (v4.2.8 Pinned)

• **Dual-Mode Operation**: Operates in offline mode without API keys (evaluation_mode = "offline_no_llm", benchmark_status = "NOT_EVALUATED") without fabricating canned scores.

• **Live DeepEval Metrics**: Evaluates Faithfulness, Answer Relevancy, Contextual Precision, and Recall using real DeepEval metric classes on LLMTestCase when OPENAI_API_KEY is configured.

• **Valuation Accuracy**: Evaluates reconciled valuation accuracy vs verified historical market bounds across benchmark scenarios in all modes.

• **True Hybrid Retrieval**: Lexical TF-IDF keyword scoring covers the full metadata-filtered candidate set before score fusion.

• **Grounded Grounding**: Reconciled FMV combines comp evidence, supervised ML regression ($R^2 = 0.9031$), and feedback calibration.

Slide 8: Feedback Loop for Continuous Improvement

Dynamic Calibration of Systemic Valuation Bias

- **Appraiser Feedback Logging**: Appraisers submit verified sales prices and 1-5 star ratings.
- **Persistent Repository**: Logged in JSON data store with error metrics.
- **Uncalibrated Bias Measurement**: Errors and systematic bias are evaluated against uncalibrated estimates (recommended / applied_factor) to prevent calibration drift.
- **Dynamic Calibration Factor**: Directly calibrated in the orchestrator reconciliation step.

Slide 9: Working Solution Demo & Local Deployment

Production-Ready Microservice & Interactive Web UI

- **Streamlit Dashboard**: 6 dedicated tabs covering Dossier, Hybrid Comps, ML Pricing, Risk, Feedback, DeepEval.
- **FastAPI Microservice**: 14 REST endpoints with full OpenAPI documentation at /docs.
- **Local Execution**: Runs locally via `uvicorn src.main:app` & `streamlit run src/frontend.py`.
- **Zero Secrets Committed**: Follows strict security hygiene.

Slide 10: Conclusion, Challenges & Q&A;

Summary of Achievements (2 Min Q&A;)

• **100% Scope Compliance**: Built all 8 Basic items and all 8 Advanced items exactly to specification.

• **Rigorous Verification**: 32 pytest unit/integration tests passing cleanly (100% success rate).

• **Complete Deliverables**: Architecture Diagram (PDF/PNG), Design Doc (PDF), API Doc (PDF), Evaluation Report (PDF), Dataset Details (PDF), Presentation Deck (PPTX/PDF).

• **Open for Panel Q&A;**: Thank you!